

Sliding Window Advertisement

Time limit: 1.00 s

Memory limit: 512 MB

A fence consists of n vertical boards. The width of each board is 1 and their heights may vary.

You want to attach a rectangular advertisement to the fence. Your task is to calculate the maximum area of such an advertisement in each window of k vertical boards, from left to right.

Input

The first line contains two integers n and k : the width of the fence and the size of the window.

After this, there are n integers $x_1, x_2, \dots, x_n$: the height of each board.

Output

Print $n - k + 1$ integers: the maximum areas of the advertisements.

Constraints

- $1 \leq k \leq n \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$

Example

Input	Output
8 3 4 1 5 3 3 2 4 1	5 6 9 6 6 4